LIST DATASTRUCTURE

1. APPEND FUNCTION

In [1]:
```
l= []
```

In [2]:
```
l.append(60)
l.append(70)
l.append(80)
l.append(90)
l.append(100)
```

In [3]:
```
l
```

Out[3]:
```
[60, 70, 80, 90, 100]
```

In [4]:
```
len(l)
```

Out[4]:
```
5
```

In [5]:
```
l.append(110)
```

In [6]:
```
l
```

Out[6]:
```
[60, 70, 80, 90, 100, 110]
```

In [8]:
```
l1=[]
```

In [14]:
```
l1 = [40,5.2,6+20j,True,'hello',[4,5,6]]
```

In [15]:
```
len(l1)
```

Out[15]:
```
6
```

In [16]:
```
print(l)
print(l1)
```

```
[60, 70, 80, 90, 100, 110]
[40, 5.2, (6+20j), True, 'hello', [4, 5, 6]]
```

In [17]:
```
print(id(l))
print(id(l1))
```

```
1909491378688
1909497859520
```

COPY FUNCTION

In [21]:
```
l2 = l.copy()
```

In [22]:
```
l2
```

Out[22]:
```
[60, 70, 80, 90, 100, 110]
```

In [23]:
```
len(l2)
```

Out[23]:
```
6
```

In [24]:
```
print(id(l2))
```

```
1909497520896
```

CLEAR FUNCTION

Clear will clear the element from the list

In [26]:
```
l2
```

Out[26]: [60, 70, 80, 90, 100, 110]

In [27]:
```
l2.clear
```

Out[27]: <function list.clear()>

COUNT FUNCTION

Count function will count the numbersin list

In [29]:
```
l.count(40)
```

Out[29]: 0

In [30]:
```
l
```

Out[30]: [60, 70, 80, 90, 100, 110]

In [32]:
```
l.count(60)
```

Out[32]: 1

In [33]:
```
l.count(90)
```

Out[33]: 1

In [34]:
```
print(l)
print(l1)
```

```
[60, 70, 80, 90, 100, 110]
[40, 5.2, (6+20j), True, 'hello', [4, 5, 6]]
```

EXTEND FUNCTION

Extend will add the both the list of numbers

In [39]:
```
x = [30,40,50]
y = [5.2,10+2j,'hello']
```

In [41]:
```
x.extend(y)
```

In [42]:
```
x
```

Out[42]:
```
[30, 40, 50, 5.2, (10+2j), 'hello', 5.2, (10+2j), 'hello']
```

In [43]:
```
l2.extend(l)
```

In [46]:
```
l2
```

Out[46]:
```
[60, 70, 80, 90, 100, 110, 60, 70, 80, 90, 100, 110]
```

In [47]:
```
print(l)
print(l1)
print(l2)
```

```
[60, 70, 80, 90, 100, 110]
[40, 5.2, (6+20j), True, 'hello', [4, 5, 6]]
[60, 70, 80, 90, 100, 110, 60, 70, 80, 90, 100, 110]
```

INDEX FUNCTION

In [60]:
```
l[3]=1000
```

In [61]:
```
l
```

Out[61]: [100, 70, 80, 1000, 100, 110]

In [62]:
```
asset=2000
```

In [63]:
```
asset
```

Out[63]: 2000

In [67]:
```
l
```

Out[67]: [100, 70, 80, 1000, 100, 110]

In [68]:
```
l[-2]
```

Out[68]: 100

In [69]:
```
l2
```

Out[69]: [60, 70, 80, 90, 100, 110, 60, 70, 80, 90, 100, 110]

INSERT FUNCTION

Insert will relace the numbers in the list of numbers

In [74]:
```
l2
```

Out[74]: [60, 70, 80, 90, 100, 110, 60, 70, 80, 90, 100, 110]

In [79]:
```
l2.insert(1,600)
```

In [81]:
```
l2
```

Out[81]: [60, 600, 70, 600, 600, 80, 90, 100, 110, 60, 70, 80, 90, 100, 110]

POP FUNCTION

POP will remove the last element from the list

In [82]:
```
l2.pop()
```
Out[82]:
```
110
```

In [83]:
```
l2.pop()
```
Out[83]:
```
100
```

In [84]:
```
l2.pop()
```
Out[84]:
```
90
```

In [85]:
```
l2.pop()
```
Out[85]:
```
80
```

In [86]:
```
l
```
Out[86]:
```
[100, 70, 80, 1000, 100, 110]
```

In [87]:
```
l.pop()
```
Out[87]:
```
110
```

In [88]:
```
l1
```
Out[88]:
```
[40, 5.2, (6+20j), True, 'hello', [4, 5, 6]]
```

In [93]:
```
l1.pop(-2)
```

Out[93]: True

In [94]:
```
l1.pop(-1)
```

Out[94]: 'hello'

Difference between pop & remove

pop will remove the last element from the list

remove will remove the particular element from the list

REMOVE FUNCTION

Remove will remove the particular element from the list

In [95]:
```
l2
```

Out[95]: [60, 600, 70, 600, 600, 80, 90, 100, 110, 60, 70]

In [100…
```
l2.remove(600)
```

In [98]:
```
l2
```

Out[98]: [60, 70, 600, 80, 90, 100, 110, 60, 70]

In [103…
```
l1
```

Out[103… [40, 5.2, (6+20j)]

In [104…
```
l1.remove(6+20j)
```

In [105…
```
l1
```
Out[105…
```
[40, 5.2]
```

REVERSE FUNCTION

Reverse will reverse the elements list

In [107…
```
l2
```
Out[107…
```
[60, 70, 80, 90, 100, 110, 60, 70]
```

In [108…
```
l2.reverse()
```

In [109…
```
l2
```
Out[109…
```
[70, 60, 110, 100, 90, 80, 70, 60]
```

In [116…
```
l1 = [100,110,120,130,140,150]
```

In [117…
```
l1.reverse()
```

In [118…
```
l1
```
Out[118…
```
[150, 140, 130, 120, 110, 100]
```

In [119…
```
l = [8.2,9+6j,'python',[9,8,7]]
```

In [120…
```
l.reverse()
```

In [121…
```
l
```
Out[121…
```
[[9, 8, 7], 'python', (9+6j), 8.2]
```

SORT FUNCTION

sort will automatically arrange in ascending order

In [122…
```
l2
```
Out[122…
```
[70, 60, 110, 100, 90, 80, 70, 60]
```
In [124…
```
l2.sort()
```
In [125…
```
l2
```
Out[125…
```
[60, 60, 70, 70, 80, 90, 100, 110]
```
In [126…
```
l2.sort(reverse=True)
```
In [127…
```
l2
```
Out[127…
```
[110, 100, 90, 80, 70, 70, 60, 60]
```
In [129…
```
l
```
Out[129…
```
[[9, 8, 7], 'python', (9+6j), 8.2]
```
In [ ]: